Knowledge Management and Collective Memory: How Teams Learn and Remember

Executive Summary

A team's knowledge exceeds what any member knows — but only if the team has systems for **creating, sharing, storing, and retrieving** knowledge collectively. Under Active Inference, collective learning is the process of updating the team's shared generative model based on joint experience. This module covers knowledge creation (Nonaka), communities of practice (Wenger), transactive memory systems, and the challenge of preserving institutional knowledge.

Learning Objectives

1. Define **collective learning** as updating the team's shared generative model
2. Apply Nonaka's **SECI model** of knowledge creation (Socialization, Externalization, Combination, Internalization)
3. Understand **transactive memory systems** — knowing who knows what
4. Design **communities of practice** that support ongoing collective learning
5. Address the challenge of **institutional memory** — preserving knowledge across personnel changes

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Key Concepts

1. Knowledge Creation: The SECI Model (Nonaka)

Knowledge flows through four transformations:

Mode	From → To	Mechanism	Example
Socialization	Tacit → Tacit	Shared experience, apprenticeship	Junior doctor learning from senior by observation
Externalization	Tacit → Explicit	Articulating know-how into documents, models	Writing a best practices guide after a successful project
Combination	Explicit → Explicit	Combining documented knowledge into new forms	Building a training curriculum from multiple process documents
Internalization	Explicit → Tacit	Learning by doing, practice	Reading the manual then doing it until it becomes second nature

2. Transactive Memory Systems

A transactive memory system is the team's knowledge of **who knows what** — a social directory of expertise:

- **Specialization**: Members develop expertise in different domains
- **Credibility**: Members trust each other's expertise
- **Coordination**: Members know who to ask for specific knowledge

Case Study — Surgical Teams (again): Research shows that surgical teams that work together regularly develop transactive memory — each member knows what the others know and don't know. Teams with strong transactive memory achieve significantly better patient outcomes than teams composed of equally skilled individuals who haven't worked together.

3. Communities of Practice (Wenger)

Informal groups that share a concern or passion and learn through regular interaction:

- **Domain**: Shared area of interest
- **Community**: Regular interaction and relationship building
- **Practice**: Shared repertoire of resources, methods, and tools

4. Institutional Memory

When experienced members leave, knowledge is lost unless the organization has:

- Documented processes (explicit knowledge)
- Mentoring relationships (tacit knowledge transfer)
- Knowledge management systems (searchable repositories)
- Overlapping team membership (redundant knowledge distribution)

Cross-References

- For organizational learning, see Organizational Systems: Learning
- For adaptive strategy, see Strategic Modeling: Learning
- For ML model retraining, see Digital Transformation: Learning

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Summary

Concept	Team/Collective Meaning
Collective learning	Updating the team's shared generative model from joint experience
SECI model	Knowledge flows between tacit and explicit across four modes
Transactive memory	The team's directory of who knows what
Communities of practice	Informal groups that learn through ongoing shared practice
Institutional memory	Preserving knowledge across personnel changes

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References

- Nonaka, I., & Takeuchi, H. (1995). *The Knowledge-Creating Company*. Oxford University Press.
- Wenger, E. (1998). *Communities of Practice*. Cambridge University Press.
- Wegner, D. M. (1987). Transactive memory. In B. Mullen & G. R. Goethals (Eds.), *Theories of Group Behavior*. Springer.